Michael Xu

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Experience

Senior Research Engineer — Sept. 2024 – Present
Microsoft Research - Deep Learning Group — *San Francisco, CA*

- Researched and scaled a novel generative flow-based RL loss function for LLM post-training, outperforming GRPO on mathematical reasoning and code generation benchmarks using veRL.
- Designed and implemented a scalable human-in-the-loop data collection platform on Xbox Insider, processing 50,000+ daily preference annotations through gamified crowdsourcing interactions with built-in quality control mechanisms, generating training data for reward modeling and RLHF alignment pipelines.

Research Engineer 2 — Feb. 2021 – Sept. 2024
Microsoft Research - Natural Language Processing — *San Francisco, CA*

- Provided technical leadership for LLM integration efforts across Xbox game studios, implementing LLM-driven systems that enhanced narrative complexity and created rich generative player interactions in multiple in-development game titles.
- Cross-functional collaboration with Microsoft AI on the Gaming Copilot Companion initiative, integrating on-device multimodal vision models with real-time gameplay analysis to provide contextual assistance across multiple game genres, resulting in improved player experience and accessibility.
- Released a dynamic branching narrative framework using LLMs, enabling procedurally generated dialogue trees that increased story variation while maintaining narrative coherence across storylines (paper).
- Developed a structured conversational prompting framework to improve LLM alignment that improved quest scenario completion rates by 25% (paper).
- Communicated gaming AI research to executive stakeholders, driving strategic decision-making and securing continued investment in AI initiatives at Microsoft Research and Xbox.

Projects

Craft an Iron Sword | *Minecraft, Conversational Agent, Tool Use, RAG* | GitHub Aug. 2021 – Present

- Pioneered an agentic AI system for Minecraft combining LLMs with composable tool calling via JavaScript execution and dynamic RAG retrieval through promise chaining; predating modern tool-use paradigms by over two years (paper).
- Extended the system with multimodal capabilities including real-time video and voice interaction; showcased by CEO Satya Nadella at Build 2024 (video) and winner of Executive Hack at Microsoft Hackathon 2022.

Education

University of Illinois at Urbana-Champaign — Champaign, IL
M.S. in Computer Science, Data Science Concentration - 3.91 GPA

Arizona State University — Tempe, AZ
B.S. in Computer Science, Minor in Statistics - 4.0 GPA

Technical Expertise

Research Areas: Reinforcement Learning & Post-Training, Reward Modeling & RLHF, LLM Alignment & Evaluation, Multimodal AI Systems, Conversational AI & NLP
Languages: Python, TypeScript / JavaScript, SQL, C++, Java
ML & Infrastructure: PyTorch, veRL, vLLM, Transformers, TensorRT, Kubernetes, Docker, Azure